

# A Sample Paper on Vector Fields

Ada Fixtura  
Alexandria Test Fixtures

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## Abstract

This is a synthetic document used only to exercise Alexandria's PDF extraction pipeline. It carries a few genuine equations so the compiled PDF embeds Computer Modern math fonts, and enough prose to make chunking meaningful.

## 1 Introduction

We consider a toy magnetostatic field so the text contains real mathematics rather than placeholder symbols. The magnetic field  $\vec{B}$  produced by a steady current follows the Biot–Savart law, and its axial component  $B_z$  can be written compactly in terms of a scalar potential  $\Phi$ . The Greek letters and operators below guarantee a nonzero math-symbol density, and the equations force the math fonts into the font resource dictionary.

## 2 A toy field equation

The field of a current loop of current  $I$  is

$$\vec{B}(\vec{r}) = \frac{\mu_0}{4\pi} \oint \frac{I d\vec{l} \times \hat{r}}{r^2}, \quad (1)$$

and along the symmetry axis the component reduces to

$$B_z = \alpha \frac{\partial \Phi}{\partial z}, \quad \alpha = \frac{\mu_0 I}{2}. \quad (2)$$

Here  $\alpha$  is a constant,  $\Phi$  is the scalar potential, and the summation  $\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$  appears purely to add a few more symbols to the page.

### 3 Discussion

The point of this fixture is not physics but coverage. A real extractor must survive a multi-page, L<sup>A</sup>T<sub>E</sub>X-generated PDF: hyphenated line breaks, ligatures such as *efficient* and *affordable*, and inline math mixed with ordinary prose. None of the numbers here mean anything; they exist so that the normalizer, the chunker, and the math-density heuristics all have something representative to chew on.

We repeat a little more prose to push the document onto a second page, because the extraction test treats this as a multi-page paper. Retrieval-augmented generation, vector databases, and keyword search over an inverted index are all mentioned here so the fixture reads like a plausible technical note rather than lorem ipsum. The conclusion, such as it is, is that vector fields make excellent excuses for embedding math fonts.